

Requirements Engineering

Software Engineering Presentation

Group members: [Name 1] • [Name 2] • [Name 3]

Estimated presentation time: 8 minutes





Introduction

- **What is Requirements Engineering?** — The disciplined process of gathering, analysing, documenting, and managing what a software system must do.
- **Simple definition:** Capture the right needs for the right users.
- **Why it matters:** Foundation for design, development, and testing; prevents rework.



Why Requirements Engineering Matters

- Reduces project failure — catches issues early
- Saves time & cost — fewer late changes
- Improves quality — clearer acceptance criteria
- Better communication — shared understanding across teams

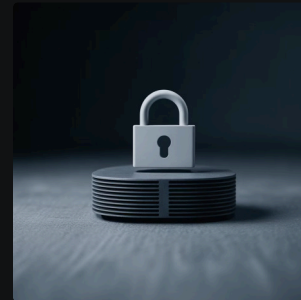
Types of Requirements



Functional Requirements

Describe specific behaviors or features

- Example: "User can add items to cart"



Non-Functional Requirements

Quality and constraints of the system

- Example: "Page loads under 2 seconds" (performance)

Requirements Engineering Process



Elicitation: collect needs from stakeholders • Analysis: resolve conflicts & prioritize • Specification: write clear, testable requirements • Validation: confirm requirements meet stakeholder needs



Real-Life Example: Online Shopping System

- **Functional:** Add to cart, checkout, process payment, order history
- **Non-Functional:** Fast (≤ 2 s pages), secure (PCI-compliant), available 99.9%
- **Design impact:** Requirements drive UI flows, security controls, and testing

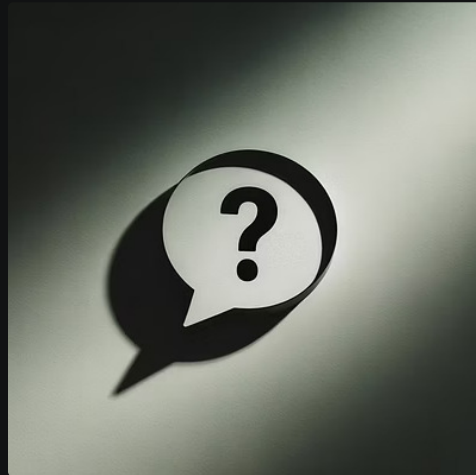


Conclusion

- Requirements Engineering (RE) is essential for successful software projects
- Clear requirements reduce risk, save time, and improve product quality
- Good RE improves user satisfaction and enables reliable testing
- Key practice: involve stakeholders early and keep requirements testable

Thank You

Any Questions?



Contact: [email@example.edu] • Course: Software Engineering